Activation function: ReLU, an example

Input	5	
ReLU	*[FILL]*	

$$\mathrm{ReLU}(x) = \max(0, x)$$

Activation function: ReLU, 1D array/tensor

Input	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
ReLU	*[FILL]*																			

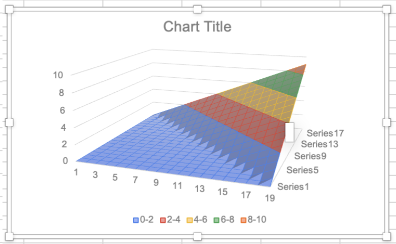
Activation function: ReLU, chart

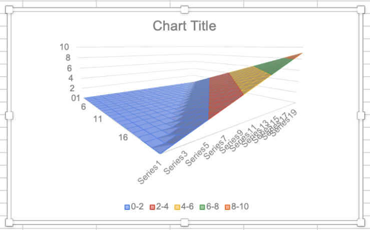
Activation function: ReLU, 2D array/tensor

Input	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
	-4.5	-9	-8.5	-8	-7.5	-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0
	-4	-8.5	-8	-7.5	-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5
	-3.5	-8	-7.5	-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1
	-3	-7.5	-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5
	-2.5	-7	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2
	-2	-6.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5
	-1.5	-6	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3
	-1	-5.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5
	-0.5	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4
	0	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
	0.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5
	1	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5
	1.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6
	2	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5
	2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7
	3	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5
	3.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8
	4	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5
	4.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9

	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
ReLU	-4.5	*[FILL]*																		

-4
-3.5
-3
-2.5
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5

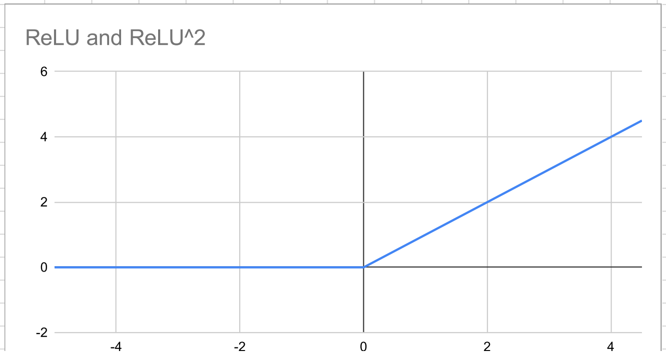


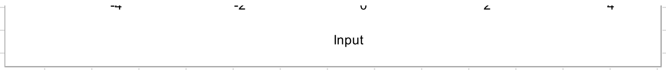


Activation function: ReLU^2

$$\mathrm{ReLU\ Squared}(x) = (\max(0, x))^2$$

Input	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
ReLU	0	0	0	0	0	0	0	0	0	0	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
ReLU^2																				



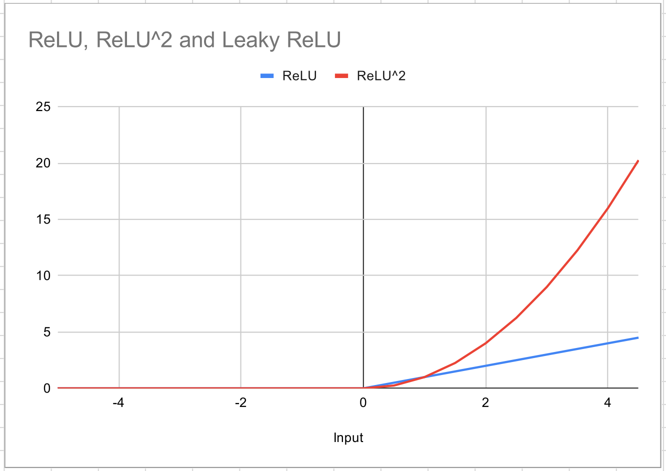


Activation function: Leaky ReLU

$$\text{Leaky ReLU}(x) = \begin{cases} x & x \geq 0 \\ \alpha x & x < 0 \end{cases}$$

Leaky alpha	0.2

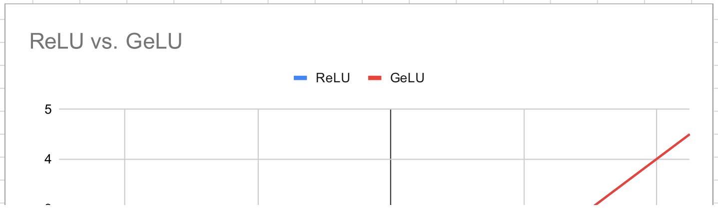
Input	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
ReLU	0	0	0	0	0	0	0	0	0	0	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
ReLU^2	0	0	0	0	0	0	0	0	0	0	0	0.25	1	2.25	4	6.25	9	12.25	16	20.25
Leaky ReLU	IF(C124>0,C124, C123*C121)																			

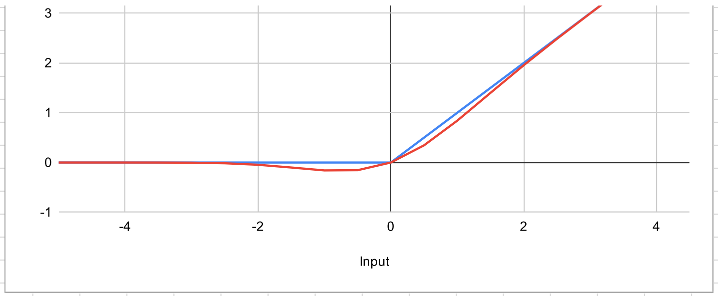


Activation function: GeLU

$$\text{GELU}(x) \approx 0.5\,x\left(1+\tanh\left(\sqrt{\tfrac{2}{\pi}}\,(x+0.044715x^3)\right)\right)$$

Input	-5	-4.5	-4	-3.5	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
ReLU	0	0	0	0	0	0	0	0	0	0	0	0.5	1	1.5	2	2.5	3	3.5	4	4.5
GeLU	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-0.1	-0.2	-0.2	0.0	0.3	0.8	1.4	2.0	2.5	3.0	3.5	4.0	4.5





Activation function: Tanh, Sigmoid, Swish, SwiGLU

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

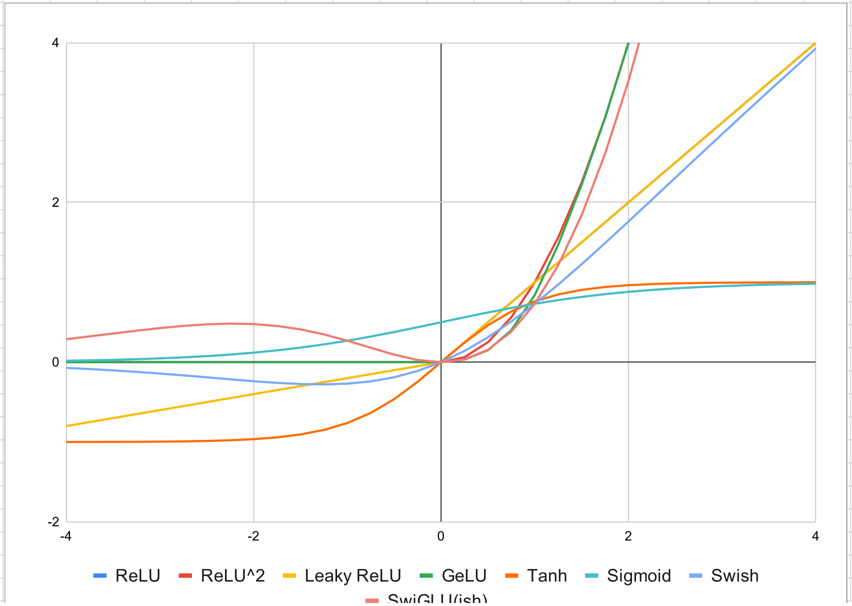
$$\mathrm{Sigmoid}(x) = \frac{1}{1 + e^{-x}}$$

$$\mathrm{SiLU}(z) = z \cdot \frac{1}{1 + e^{-z}}$$

$$\mathrm{SwiGLU}(x) = (xW_1) \odot \mathrm{SiLU}(xW_2)$$

Input	-4	-3.75	-3.5	-3.25	-3	-2.75	-2.5	-2.25	-2	-1.75	-1.5	-1.25	-1	-0.75	-0.5	-0.25	0
ReLU	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ReLU^2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Leaky ReLU	-0.8	-0.75	-0.7	-0.65	-0.6	-0.55	-0.5	-0.45	-0.4	-0.35	-0.3	-0.25	-0.2	-0.15	-0.1	-0.05	0
GeLU	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Tanh	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0	-1.0	-0.9	-0.9	-0.8	-0.8	-0.6	-0.5	-0.2	0.0
Sigmoid	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.3	0.3	0.4	0.4	0.5
Swish	-0.1	-0.1	-0.1	-0.1	-0.1	-0.2	-0.2	-0.2	-0.2	-0.3	-0.3	-0.3	-0.3	-0.2	-0.2	-0.1	0.0
SwiGLU(ish)	0.3	0.3	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0.4	0.3	0.3	0.2	0.1	0.0	0.0

Input	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3	3.25	3.5	3.75	4
ReLU	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3	3.25	3.5	3.75	4
ReLU^2	0.0625	0.25	0.5625	1	1.5625	2.25	3.0625	4	5.0625	6.25	7.5625	9	10.5625	12.25	14.0625	16
Leaky ReLU	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3	3.25	3.5	3.75	4
GeLU	0.0	0.1	0.4	0.8	1.5	2.2	3.1	4.0	5.1	6.2	7.6	9.0	10.6	12.3	14.1	16.0
Tanh	0.2	0.5	0.6	0.8	0.8	0.9	0.9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Sigmoid	0.6	0.6	0.7	0.7	0.8	0.8	0.9	0.9	0.9	0.9	0.9	1.0	1.0	1.0	1.0	1.0
Swish	0.1	0.3	0.5	0.7	1.0	1.2	1.5	1.8	2.0	2.3	2.6	2.9	3.1	3.4	3.7	3.9
SwiGLU(ish)	0.0	0.2	0.4	0.7	1.2	1.8	2.6	3.5	4.6	5.8	7.1	8.6	10.2	11.9	13.7	15.7



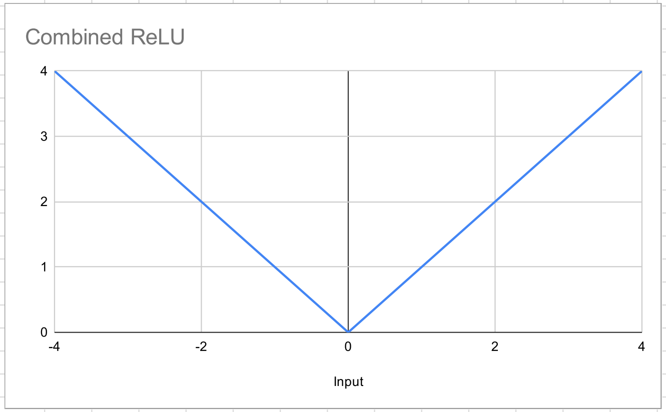
[Exercise] Where do activation functions agree?

Where do they diverge?

Which is the most interesting quadrant here?

Activation function: Combined ReLU

Input	-4	-3.75	-3.5	-3.25	-3	-2.75	-2.5	-2.25	-2	-1.75	-1.5	-1.25	-1	-0.75	-0.5	-0.25	0	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3	3.25	3.5	3.75	4
ReLU	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3	3.25	3.5	3.75	4
ReLU(-x)	4	3.75	3.5	3.25	3	2.75	2.5	2.25	2	1.75	1.5	1.25	1	0.75	0.5	0.25	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ReLU+-Re	4	3.75	3.5	3.25	3	2.75	2.5	2.25	2	1.75	1.5	1.25	1	0.75	0.5	0.25	0	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3	3.25	3.5	3.75	4



[Exercise] why would combining activation functions be useful?